CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

November 2013

CL421 Advanced Structural Analysis

Date: 23.11.2013, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. All required sketches/data are available at the end of the paper.

SECTION – I

Q - 1 Draw the neat sketch of suspension bridge and show the components of bridge. Also mention the forces acting on different components due to vehicular load. [07]

Q - 2 Calculate design bending moment and design shear force for following data of concrete deck slab. [14]

Clear span – 9 m

Clear width of roadway – 7.5 m

Footpath – 600 mm wide on either side

Wearing coat – 100 mm thick

Width of bearing = 400 mm

Loading – IRC Class AA tracked vehicle

Material – Concrete grade M 40 & Fe 500 grade HYSD bars

Assume the thickness of slab to be 50 mm per meter span of the slab.

OR

Q - 2 Calculate design bending moment and design shear force for following data of deck slab. [14]

Clear span – 5 m

Width of footpath – 1000 mm on either side

Wearing coat – 80 mm

Loading – IRC class A

Material – Concrete grade M 25 & Fe 415 grade HYSD bars

Q - 3 Attempt any **Two:** [14]

(i) Explain Corbon's Method for the analysis of a typical tee beam and slab deck bridge.

(ii) Obtain the values of short span and long span bending moments in case of an interior panel of a T-beam bridge having the following details:

Dimension of the panel – 3 m x 3.5 m

Loading – IRC Class A

Loading pattern- Two wheels (each of 57 kN) adjusted symmetrically with respect to centre of the panel

(iii) Obtain Courbon's reaction factor and the maximum bending moment in case of a T-beam bridge having the following details:

Roadway – 2 lanes

Loading – IRC Class A

No. of main girders – 3, c/c spacing = 2.6 m

Span of the bridge – 16 m

Kerb width – 600 mm on either side

SECTION – II

Q - 4 Explain material and geometric nonlinearity. Also give assumptions for nonlinear analysis. **[07]**

Q - 5 (a) Calculate the displacement with given data using Modified Newton Raphson method (minimum three iteration).(Ref. Figure 1) **[08]**

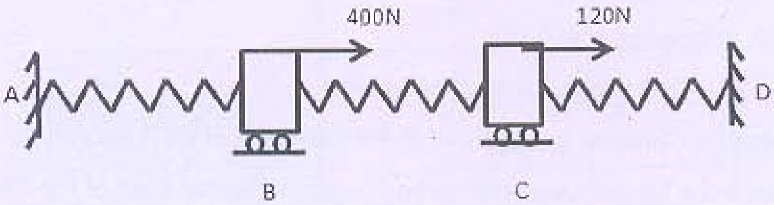

Figure: 1

Secant stiffness matrix (UA and UD=0)

$$\begin{Bmatrix} 400 \\ 120 \end{Bmatrix} = \begin{vmatrix} 250 + 5U_B - 15U_C & -150 - 15U_B + 15U_C \\ -150 - 15U_B + 15U_C & 350 + 15U_B + 5U_C \end{vmatrix} \begin{Bmatrix} U_B \\ U_C \end{Bmatrix}$$

Tangent stiffness matrix

$$\begin{Bmatrix} 400 \\ 120 \end{Bmatrix} = \begin{vmatrix} 250 + 10U_B - 30U_C & -150 - 30U_B + 30U_C \\ -150 - 30U_B + 30U_C & 350 + 30U_B + 10U_C \end{vmatrix} \begin{Bmatrix} dU_B \\ dU_C \end{Bmatrix}$$

(b) Define Dome. **[06]**

Explain Meridional thrust and hoop force with neat sketch.

OR

(b) (i) Enlist uses of dome. [02]

(ii) Explain various types of dome. [04]

Q - 6 Attempt any **Two:** [14]

(i) Derive an expression for Meridional thrust and hoop force for a conical dome subjected to uniformly distributed load at crown.

(ii) Design a conical roof for following requirements:

Diameter of hall = 16 m ; Rise of dome = 4 m ; Live Load = 2 kN/m^2

Use M 20 grade concrete and Fe 415 grade steel.

(iii) A spherical dome has following details:

Rise of dome = 4 m

Diameter of hall = 12 m

Live Load = 1.5 kN/m^2

A lantern is provided at its top which causes a dead load of 22 kN.

Calculate Meridional stress at various locations at 10° interval starting from 1° to the base of dome due to combined load.

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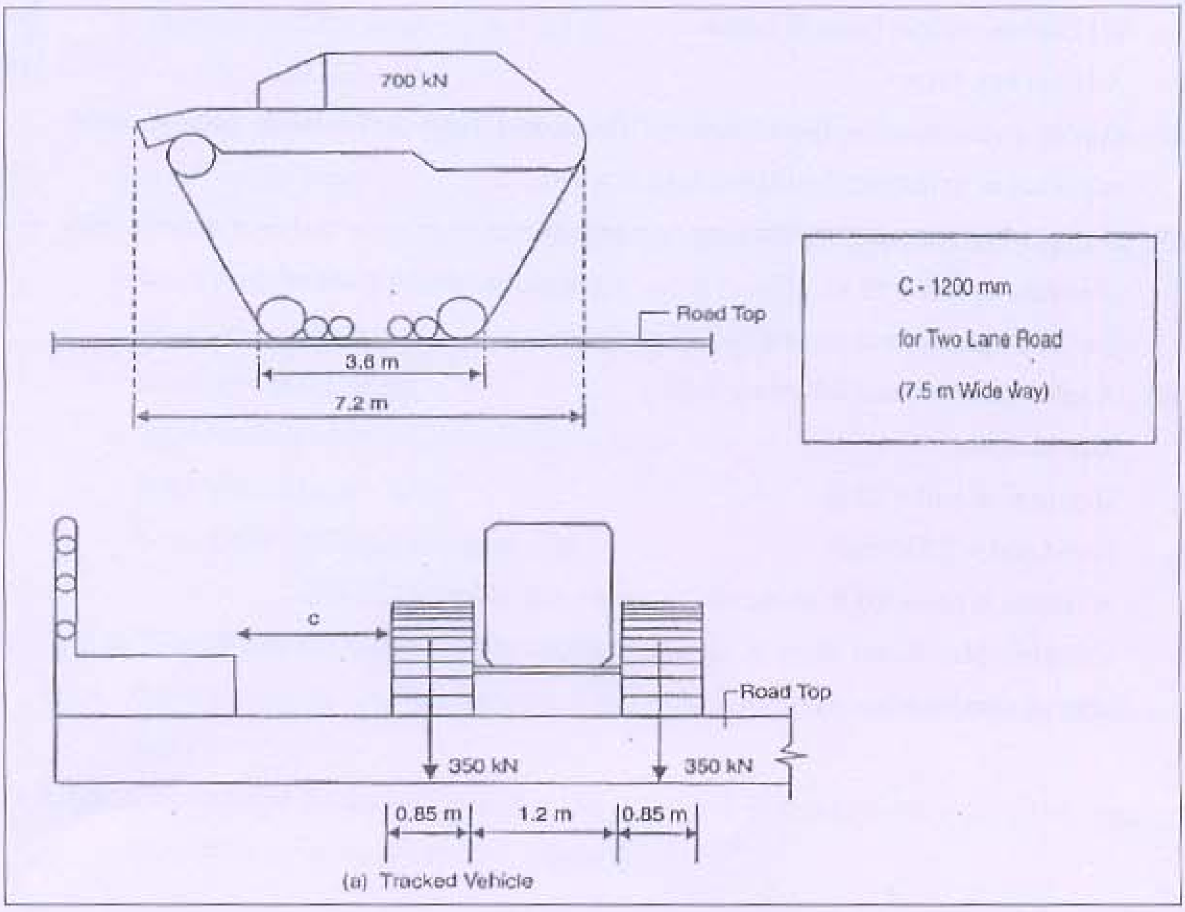


Figure: 1 Class AA (Tracked)

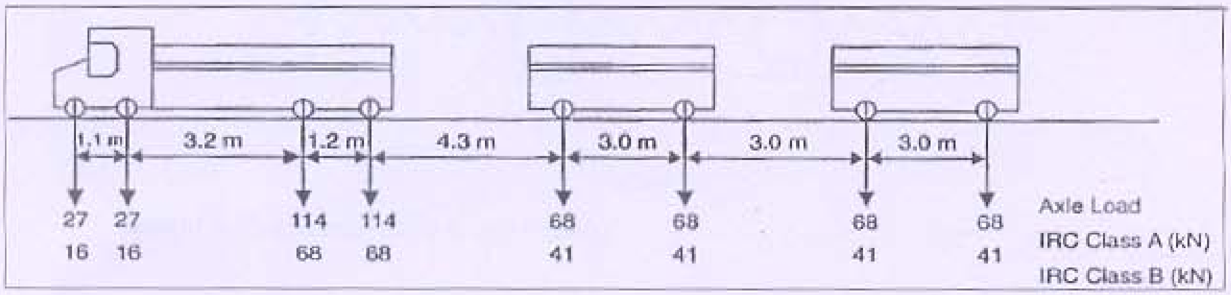


Figure: 2 Side View Details (IRC Class A & B)

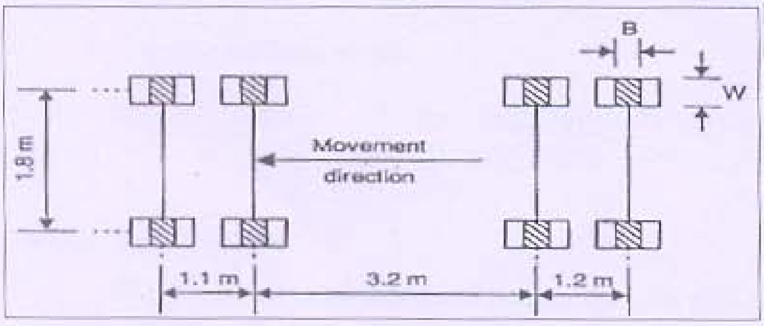


Figure: 3 Wheel impressions on road (IRC Class A & B)

Axle load	Ground contact area	
	B (mm)	W (mm)
114	250	500
68	200	380
41	150	300
27	150	200
16	125	175

Table: 1

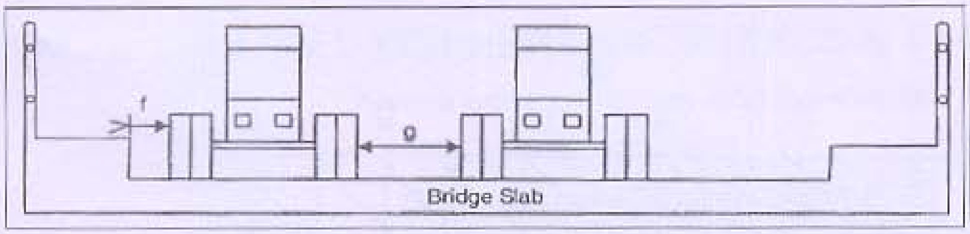


Figure: 4 Transverse placement of wheels (IRC Class A & B)

$f = 150$ mm
$g = 1200$ mm
(for 7.5 m wide way)

Figure: 5

$$I_f = \frac{A}{B + L}$$

where
I_f = impact factor
A = constant, 4.5 for RCC bridges, 9.0 for steel bridges
B = constant, 6.00 for RCC bridges, 13.50 for steel bridges
L = span in m.

Figure: 6 Impact value for IRC Class A or Class B

For IRC Class AA and 70R loading

1. Spans < 9 m
   (a) *Tracked vehicle.* 25% for spans up to 5 m linearly reducing to 10% for spans up to 9 m.
   (b) *Wheeled vehicle.* 25% for spans up to 9 m.
2. Spans > 9 m
   (a) *Tracked vehicle.* For RCC bridges, 10% for spans up to 40 m and as per graph for spans > 40 m. For steel bridges, 10% for all spans.
   (b) *Wheeled vehicle.* For RCC bridges, 25% for spans up to 12 m, and in accordance with graph for spans > 12 m. For steel bridges, 25% for spans up to 23 m and as per graph (IRC 6) for spans exceeding 23 m.

Appropriate impact factors as mentioned below need be considered for substructures as well.

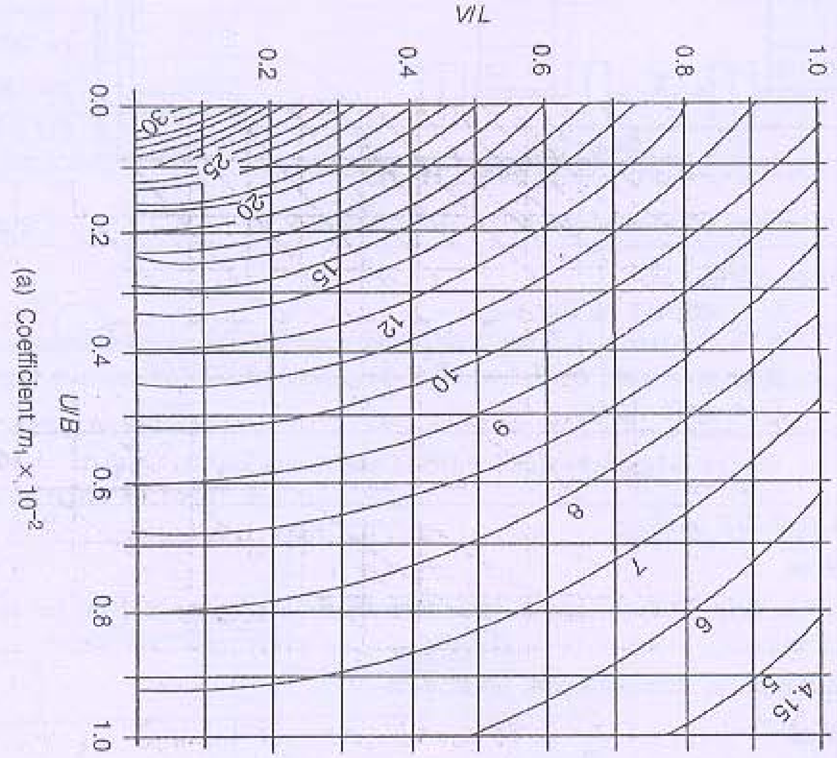
- At the bottom of the bed block: 0.5
- For the top 3 m of the substructure: 0.5 to 0.0
- For portion of the substructure > 3 m below the block: 0.0

Figure: 7 Impact value for IRC Class AA or Class 70 R loading

B/l	*α for sss**	*α for cs**	*B/l*	*α for sss*	*α for cs*
0.1	0.40	0.40	1.1	2.60	2.28
0.2	0.80	0.80	1.2	2.64	2.36
0.3	1.16	1.16	1.3	2.72	2.40
0.4	1.48	1.44	1.4	2.80	2.48
0.5	1.72	1.68	1.5	2.84	2.48
0.6	1.96	1.84	1.6	2.88	2.52
0.7	2.12	1.96	1.7	2.92	2.56
0.8	2.24	2.08	1.8	2.96	2.60
0.9	2.36	2.16	1.9	3.00	2.60
1.0	2.48	2.24	2.0 and above	3.00	2.60

* sss = simply supported slab, cs = continuous slab

Table: 2 Values of α for slabs (IRC 21)



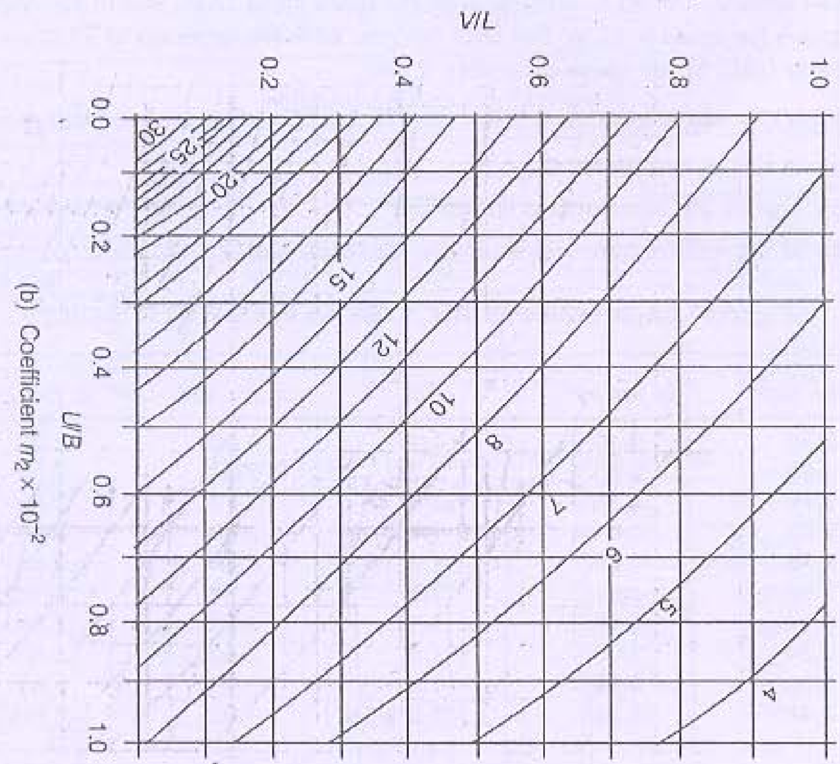


Figure 8: Moment coefficient m_1 and m_2 for $K = 0.9$